

Question 1: Two-Room Vacuum World

(a) PEAS Description

Performance Measure: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take $p = 1$ unless specified).

Environment: Two-room grid consisting of square A (left) and square B (right). Each square can be Clean or Dirty.

Actuators: Suck, Right, Left, NoOp.

Sensors: Location sensor (perceives whether in square A or B) and dirt sensor (perceives whether current square is Dirty or Clean).

(b) Trace for Agent X (Simple Reflex Agent)

Agent Rule: If current square is Dirty, action is Suck; otherwise move to the other square.

Initial State: Agent at square A; Square A = Dirty, Square B = Dirty. $p = 1$.

Time Step(t):	Location	Status (A, B)	Action	New Status (A, B)	Clean Squares at end of step	Move Penalty	Step Score
1	A	(Dirty, Dirty)	Suck	(Clean, Dirty)	1	0	$+1 - 0 = +1$
2	A	(Clean, Dirty)	Right	(Clean, Dirty)	1	1	$+1 - 1 = 0$
3	B	(Clean, Dirty)	Suck	(Clean, Clean)	2	0	$+2 - 0 = +2$
4	B	(Clean, Clean)	Left	(Clean, Clean)	2	1	$+2 - 1 = +1$
5	A	(Clean, Clean)	Right	(Clean, Clean)	2	1	$+2 - 1 = +1$
6	A	(Clean, Clean)	Left	(Clean, Clean)	2	0	$+2 - 0 = +2$

(c) Trace for Agent Y (Reflex Agent with Internal State)

Agent Rule: If current square is Dirty, action is Suck; otherwise, if it remembers both squares are clean, action is NoOp; otherwise move to the other square.

Initial State: Agent at square A; Square A = Dirty, Square B = Dirty. Internal state starts with both squares unverified/dirty. $p = 1$.

Time Step (t):	Location	Status (A, B)	Internal State	Memory	Action	New Status (A, B)	Clean Squares at end of step	Move Penalty	Step Score
1	A	(Dirty, Dirty)	A: Cleaned, B: ?		Suck	(Clean, Dirty)	1	0	$+1 - 0 = +1$
2	A	(Clean, Dirty)	A: Cleaned, B: ?		Right	(Clean, Dirty)	1	1	$+1 - 1 = 0$
3	B	(Clean, Dirty)	A: Cleaned, B: Cleaned		Suck	(Clean, Clean)	2	0	$+2 - 0 = +2$
4	B	(Clean, Clean)	Both Cleaned		NoOp	(Clean, Clean)	2	0	$+2 - 0 = +2$
5	B	(Clean, Clean)	Both Cleaned		NoOp	(Clean, Clean)	2	0	$+2 - 0 = +2$
6	B	(Clean, Clean)	Both Cleaned		NoOp	(Clean, Clean)	2	0	$+2 - 0 = +2$

(d) Score Functions & Comparison

Agent X Score Function:

$$\text{score}_X = 10 - 4p$$

Agent Y score Function:

source $yp=10-p$

condition of y strictly better than x

Agent Y is strictly better than Agent X for any positive movement penalty

(e) Rationality Interpretation

Which Agent is Rational? Agent Y is rational under this performance measure.

Explanation: Rationality depends on the expected value of the performance measure given the percept sequence to date and built-in knowledge. Since moving incurs a penalty ($p > 0$), once all rooms are cleaned, any subsequent movement reduces utility unnecessarily. Agent X lacks internal memory, so it oscillates between rooms infinitely and accumulates movement penalties. Agent Y maintains state to track whether both rooms are clean, enabling it to select NoOp and maximize its total reward. Even though both agents receive identical percepts at step 4 ([B, Clean]), Agent Y utilizes percept history stored in its internal state to make the rational decision.

Question 2: Task Environment Analysis and Agent Selection

(a) Classification and Justifications

E1: Autonomous Taxi Driving

Observability: Partially observable (cannot see around corners or inside other cars).

Number of agents: Multi-agent (other drivers/pedestrians act as agents).

Determinism: Stochastic (traffic, mechanical variations, weather are unpredictable).

Episodic/Sequential: Sequential (current driving decisions affect future safety and positions).

Static/Dynamic: Dynamic (environment changes continuously while deciding).

Discrete/Continuous: Continuous (steering angle, velocity, and time are continuous).

E2: Part-Picking Robot

Observability: Fully observable (camera observes the picking area on the conveyor belt).

Number of agents: Single agent (robot operates independently).

Determinism: Stochastic (part arrival times and belt speeds contain physical variance).

Episodic/Sequential: Episodic (picking one part does not change how the next independent part is picked).

Static/Dynamic: Semi-dynamic (conveyor belt moves continuously, but internal state score doesn't decay).

Discrete/Continuous: Continuous (robotic arm movements and conveyor locations are continuous).

E3: Chess-Playing Agent (with clock)

Observability: Fully observable (entire board state is visible).

Number of agents: Multi-agent (competing against an opponent).

Determinism: Deterministic (moves strictly dictate outcome with no randomness).

Episodic/Sequential: Sequential (early moves determine late game positions).

Static/Dynamic: Semi-dynamic (board doesn't change during thinking, but chess clock ticks down).

Discrete/Continuous: Discrete (finite set of board positions and legal moves).

(b) Difficulty Scores and Ranking

Environment ,ObservabilityAgents DeterminismEpisodic/SeqStatic/DynamicDiscrete/
ContTotal Score

E1 (Autonomous Taxi) 11111.016.0
E2 (Part-Picking Robot) 00100.512.5
E3 (Chess with Clock) 010

$\{E1 \setminus (6.0) > E2 \setminus (2.5) = E3 \setminus (2.5)\}$

(c) Suitable Agent Architectures

Hardest Environment (E1 - Autonomous Taxi): Utility-Based Agent / Learning Agent.

Justification: Navigating dynamic traffic under uncertainty requires weighing complex tradeoffs (speed vs. safety vs. passenger comfort) using a continuous utility function, while learning allows adaptation to varying road conditions.

Easiest Environment (E2 / E3):

For E2 (Part-Picking Robot): Model-Based Reflex Agent.

Justification: It handles continuous visual input and belt movements by maintaining an internal model of part trajectory, while remaining computationally lightweight.

For E3 (Chess): Goal-Based / Utility-Based Agent.

Justification: Chess requires look-ahead search and planning toward a goal state (checkmate) or evaluating board state utility.

(d) Impact of Environment Properties on Agent Program Complexity

Environment properties directly govern the internal state requirements and decision algorithm of the agent:

Partial Observability & Dynamic nature force the agent to maintain an internal state (memory) and keep track of unobserved dynamics over time.

Sequential & Stochastic properties prevent simple condition-action lookups; they demand planning ahead across non-deterministic outcome trees.

Continuous domain mandates real-time mathematical optimization and feedback loops rather than discrete search trees.

Question 3: Turing Test Data Across the History of AI

(a) Correct-Identification Rate and Fooling Rate

$\{\text{Correct-Identification Rate}\} = \{\text{Correctly Identified}\} \{ \text{Interrogations} \} 100\%$

ELIZA

$\{\text{Correct-ID Rate}\} = \{47\} \{50\} = 94\% \{\text{Fooling Rate}\} = 6\%$

chat bot B

{Correct-ID Rate} = {36}{50} = 72%
{Fooling Rate} = 28%

system c :

{Correct-ID Rate} = {29}{50} = 58%{Fooling Rate} = 42%

(b) Turing's 70% Criterion Assessment

Turing's Criterion: Turing predicted interrogators would have no more than a 70% chance of making the correct identification (i.e., Correct-ID Rate $\leq 70\%$, or Fooling Rate $\geq 30\%$).

Pure Guessing Baseline: 50%.

ELIZA (94% Correct ID): Fails criterion. It is 44% higher than the random guessing level (50%).

Chatbot B (72% Correct ID): Fails criterion (72% > 70%). It is 22% higher than random guessing.

System C (58% Correct ID): Satisfies criterion (58% $\leq 70\%$). It is only 8% above the pure random guessing level.

(c) Improvement in Fooling Rate

ELIZA to System C:

Percentage Points Improvement: 42% - 6% = {36 { percentage points}}

As a Ratio:

46%/6% = 7.0

2. chatbot b to system c :

Percentage Points Improvement: 42 - 28 = 14

(d) Interpretation: Turing Test vs. Genuine Thinking / Rationality

The Turing Test evaluates acting humanly, not thinking humanly or acting rationally.

A high fooling rate shows that System C excels at imitating conversational cues, surface syntax, and human-like dialogue patterns.

According to Searle's Chinese Room Argument, executing symbol manipulation that deceives an observer does not imply understanding or intentionality. System C processes statistical patterns over text without semantic awareness or rational decision-making grounded in logic and performance optimization.

Question 4: Rational Route Choice under Uncertainty

Given Formula & Data

Utility Function

$U = 100 - T - 20 \cdot L$

where T is travel time in minutes, and $L = 1$ if $T > 55$

otherwise $L = 0$

(a) Utility for Each Route in Each Outcome

Route R1 (Highway):

Normal: $T = 30 \text{ min}$ ($55 \leq T \implies L = 0$)

Disrupted: $T = 70 \text{ min}$ ($T > 55 \implies L = 1$, LATE TRIP)

Route R2 (City road):

Normal: $T = 45 \text{ min}$ ($55 \leq T \implies L = 0$)

Disrupted: $T = 60 \text{ min}$ ($T > 55 \implies L = 1$, LATE TRIP)

Route R3 (Ring road):

Normal: $T = 50 \text{ min}$ ($55 \leq T \implies L = 0$)

Disrupted: $T = 50 \text{ min}$ ($55 \leq T \implies L = 0$)

Route Normal Utility Disrupted Utility Late Outcome Indicator ($L=1$)

R1 70 -10 Disrupted (70 min)

R2 55 20 Disrupted (60 min)

R3 50 50 None

(b) Expected Utility and Decision

Expected Utility formula: $E[U] = P(\text{normal}) U(\text{normal}) + P(\text{disrupted}) U(\text{disrupted})$

$E[U(R1)]: 46$

$E[U(R2)]: 51.5$

$E[U(R3)]: 50$

Decision: The rational agent chooses Route R2 (City road) because it maximizes expected utility (51.5).

(c) Re-evaluation under New Percept (Radio Alert)

Updated probability of disruption on R1: $P(\text{disrupted}) = 0.80$, so $P(\text{normal}) = 0.20$.

Recomputed $E[U(R1)]: 6$

Re-decision: Comparing $E[U(R1)] = 6$, $E[U(R2)] = 51.5$, $E[U(R3)] = 50$:

The agent remains on Route R2 (or switches to R2 if it was previously considering R1).

(d) Actual Outcome vs. Omniscient Agent

The agent chose Route R2 in part (b), and disruption occurred.

Actual Utility Obtained: $U(\text{Route R2, disrupted}) = 20$.

Was the agent irrational? No. Rationality depends on expected performance based on prior information at decision time, not on actual realized outcomes after the fact.

Rational vs. Omniscient Agent:

An omniscient agent knows the exact future outcome in advance and would choose R3 (Utility = 50).

A rational agent chooses the action that maximizes expected performance given incomplete information.

(e) Interpretation of Rationality

Rationality is a function of the percept sequence to date and the performance measure. In part (c), the physical roads did not change, but the updated percept sequence (the radio alert) provided new information that altered the probability distribution of outcomes. Because rational choices maximize expected utility given current knowledge, changing information updates the expected utility calculation and leads to different decision choices